



**CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)**

BSc PHYSICS HONOURS

Programme	B.Sc. Physics Honours				
Course Title	Electronics III				
Type of Course	Core in Major				
Semester	VII				
Academic Level	400 - 499				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	PHY2CJ101- Electronics I and PHY6CJ305- Electronics II				
Course Summary	Exploration of cutting-edge concepts and methodologies in digital and analog electronics, delving into advanced topics such as high-frequency circuit design, mixed-signal systems, and emerging semiconductor technologies.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understand the fundamental principles of analog and digital electronics.	Understand	Basic Concepts	Quizzes, Tests

CO2	Analyse different types of amplifiers and their applications.	Analyse	Applications	Homework Assignments
CO3	Design amplifier circuits based on given specifications.	Apply	Circuit Design	Laboratory Experiments
CO4	Analyse the operation of different types of FETs (JFETs, MOSFETs).	Analyse	Device Operation	Homework Assignments
CO5	Understand the operational principles of Operational Amplifiers (Op-amps).	Understand	Basic Concepts	Quizzes, Assignments
CO6	Analyse and design sequential logic circuits using state diagrams and flip-flops.	Analyse	Circuit Design	Laboratory Experiments
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (45 +30)	Marks (70)
I	BJT AND FET FREQUENCY RESPONSE		12	18
	1	Decibels and general frequency considerations	3	
	2	Low frequency analysis: BJT and FET amplifiers - Bode plots	4	
	3	High frequency response – Miller effect capacitance	3	
	4	Multistage frequency effects and square wave testing	2	
	Sections 11.3 – 11.11 of Book1			

II	APPLICATIONS OF OPERATIONAL AMPLIFIERS		10	16
	6	Operational amplifier frequency responses - bode plot analysis	2	
	7	Filters - active low pass, high pass and band pass Butterworth filters, band pass filter with multiple feedback, notch filter.	3	
	8	Oscillators - Wien bridge oscillator, Astable and monostable multivibrators, Schmitt triggers	2	
	9	OPAMP as inverter, scale changer, summer, V to I converter	1	
	10	Integrator and Differentiator	2	
	Relevant Sections from Book 2			
III	DIGITAL ELECTRONICS		18	23
	12	Minimization of Boolean functions using Karnaugh map and representation using logic gates	4	
	13	Flip flops and registers: JK and MS JK and D flip-flops, shift registers using D and JK flip flops and their operations	5	
	14	Counters: shift registers as counters, ring counter, design of synchronous and asynchronous counters, state diagram, cascade counters	6	
	15	Memory: basic idea of static and dynamic RAM, basics of charge coupled devices	2	
	16	R-2R ladder D/A converter	1	
	Relevant Sections from Book 3			
IV	MICROPROCESSORS AND MICROCONTROLLERS		5	13
	17	Introduction to 8 bit microprocessor, internal architecture of Intel 8085 register organisation	2	

	18	Microcontrollers and embedded systems	1	
	19	AVR architecture: General purpose registers and data memory (no coding required)	2	
	Relevant Sections from Book 4			
V	PRACTICALS		30	
	Conduct any 6 experiments from the given list and 1 additional experiment, decided by the teacher-in-charge, related to the content of the course. The 7 th experiment may also be selected from the given list. Other experiments listed here may be used as demonstrations of the concepts taught in the course.			
	1	Design and construct OPAMP based summing and averaging amplifiers for three suitable inputs. Compare the designed and observed outputs.		
	2	Design and construct an astable multivibrator using OPAMP for suitable frequencies.		
	3	Design and construct a monostable multivibrator using OPAMP for suitable pulse widths.		
	4	Design and construct OPAMP based precision half and full wave rectifiers. Observe the o/p on CRO and study the circuit operation.		
	5	Design and construct a voltage controlled oscillator using timer IC 555. Study the performance.		
	6	Design and construct a narrow band-pass filter for a given centre frequency using a single OPAMP with multiple feedback. Study the frequency response.		
	7	Design and construct a two stage I.F amplifier circuit. Study the frequency response of single and coupled stages.		

8	Design and construct a differential amplifier using transistors. Study frequency response and measure i/p, o/p impedances. Also measure CMRR of the circuit.		
9	Design and construct a d.c voltage regulator using transistors and Zener diodes. Study the line and load regulation characteristics for suitable o/p voltage and maximum load current.		
10	Design and construct a Wien bridge oscillator using OPAMP for different frequencies. Compare designed and observed frequencies.		
11	Design and construct a triangular wave generator using OPAMPs for different frequencies.		
12	Design and construct Schmidt triggers using OPAMPs - for symmetrical and non-symmetrical LTP/UTP. Trace hysteresis curve.		
13	4 bit D/A converter using R-2R ladder network. Realization of 4 bit A/D converter using D/A converter.		
14	Design and construct a 3 bit binary to decimal decoder using suitable logic gates. Verify the operation.		
15	Study of 4 bit binary counter (IC 7493) and 4 bit decade counter (IC 7490) in various modes. Use the counters as frequency dividers.		
16	Set up a four bit shift register IC 7495 and verify right shift and left shift operations for different data inputs.		
17	Design and construct Second order Butterworth Low pass, High Pass and Band Pass filters using OPAMPs. Study the performance in each case.		
18	a). Design and construct OPAMP based circuit for solving a second order differential equation. Study the performance. b). Design and construct OPAMP based circuit for solving a simultaneous equation. Study the performance.		

19	Design and construct a Darlington pair amplifier using medium power transistors for a suitable output current. Study the frequency response of the circuit and measure the i/p and o/p impedances.		
20	a) Study the V-I characteristics of a JFET. Determine pinch-off voltage, saturation drain current and cut-off voltage of the device. b) Design and construct a low frequency common source amplifier using JFET. Study the frequency response, measure the i/p and o/p impedances.		

Books and References:

1. Electronic devices and circuit theory by Robert Boylestad and Louis Nashelsky (Book 1)
2. OPAMPS and Linear Integrated Circuits by Ramakant A. Gaykwad (Book 2)
3. Fundamentals of Microprocessors and Microcomputers by B. Ram (Book 3)
4. The AVR microcontroller and embedded systems using assembly and C (Book 4)
5. Electronics Lab Manual Vol 1 and 2 by K. A. Navas

Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO 4	PSO 5	PSO 6	PO 1	PO2	PO3	PO 4	PO 5	PO 6	PO 7
CO 1	3	1	0	0	3	1	2	1	1	0	3	1	0
CO 2	3	2	2	0	2	1	2	1	1	0	3	1	0
CO 3	3	2	3	1	2	1	2	1	2	0	3	1	0
CO 4	3	2	2	0	2	1	2	1	1	1	3	1	0
CO 5	3	1	1	0	3	1	2	1	1	0	3	1	0
CO 6	3	2	3	1	1	2	2	1	2	0	3	1	0

Correlation Levels:

Level	Correlation
0	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Discussion / Seminar
- InternalTheory / Practical Exam
- Assignments / Viva
- End Semester Exam (70%)

Mapping of COs to Assessment Rubrics

	Internal Theory / Practical Exam	Assignment / Viva	Practical Skill Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓
CO 6		✓	✓	